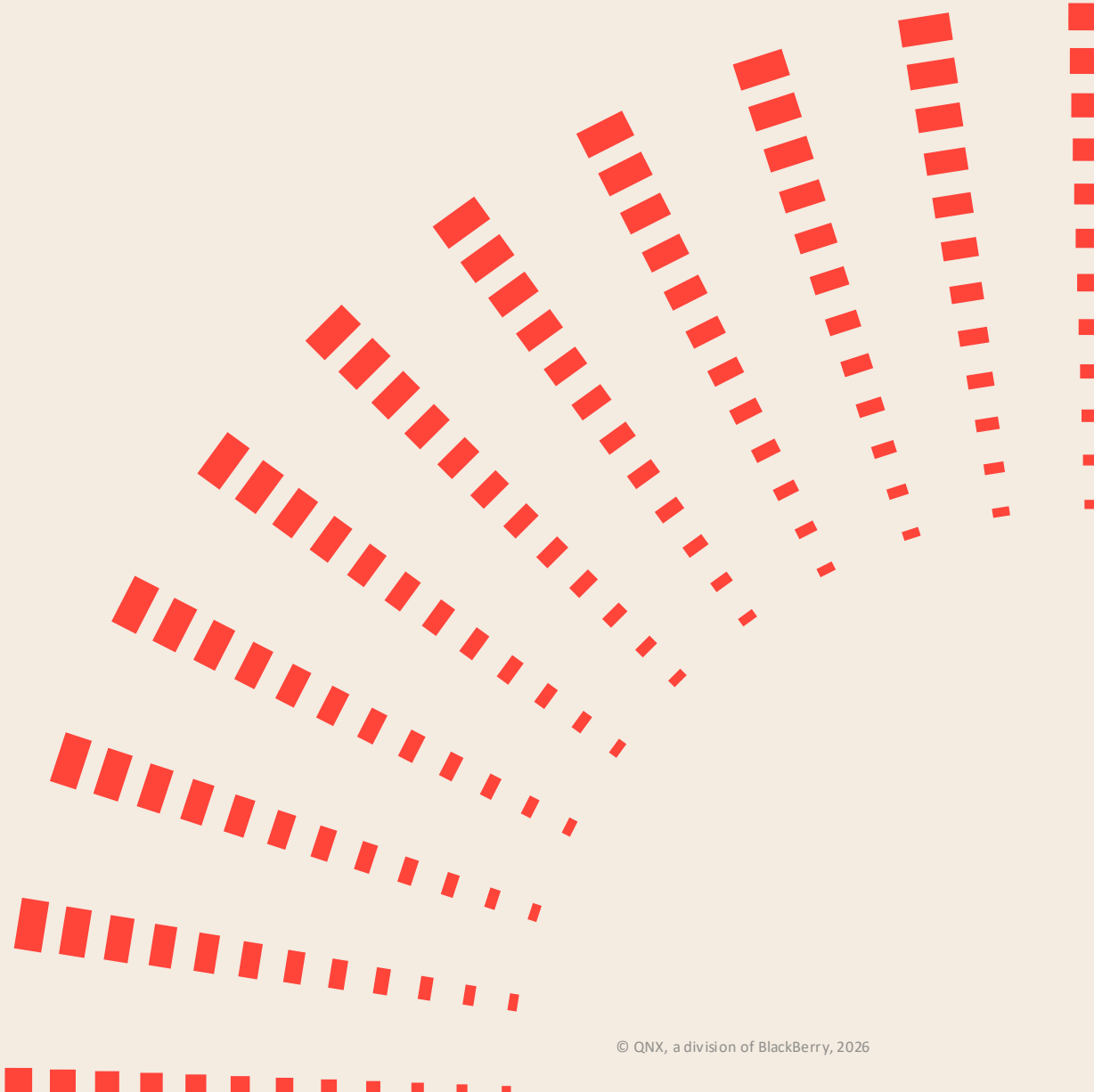
QNX on Raspberry Pi

Basma Ben Taleb, Product Management Student



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QNX Everywhere on Raspberry Pi

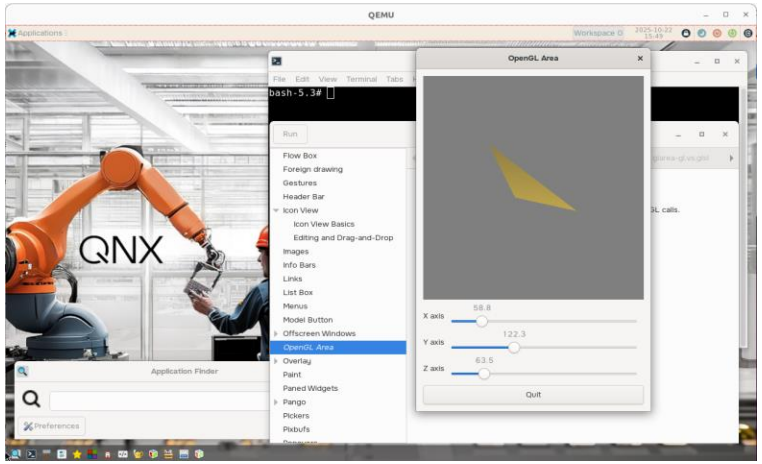
Quick Start Target Image:

- no playing with BSPs, build scripts, configurations, policies, etc
- works out of the box and includes some basic samples

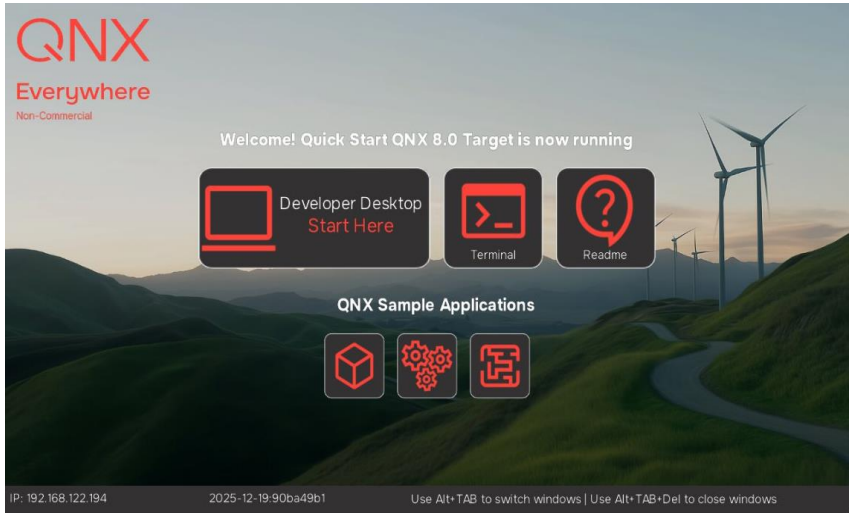
Custom Target Image:

- uses the BSP, mkqnximage, snippets, scripts
- builds a fully custom image with exactly what you want in it

Released recently: QNX self-hosted desktop!



We've also open-sourced our BSPs, for fully customized builds



Web search:

"QNX Everywhere Documentation"

www.qnx.com/developers/docs/qnxeverywhere/index.html

Next up: Pi distribution...

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Your Raspberry Pi 5 Kit!

Password for VNC is qnxuser

Kit includes:

- Raspberry Pi 4 or 5 and accessories (case, power, heatsink, fans)
- Micro-SD card with image of QNX 8.0 for Pi (already loaded in the Pi)
- Your Micro-SD card is specific to you! Your micro-SD card came with a hostname – note it!
- A Logitech C270 HD Webcam

Here's the plan:

- Connect to the QNXLaptops Wi-Fi network (password: everywhere)
- Download a VNC Client (https://sourceforge.net/projects/tigervnc/files/stable/1.16.2/)

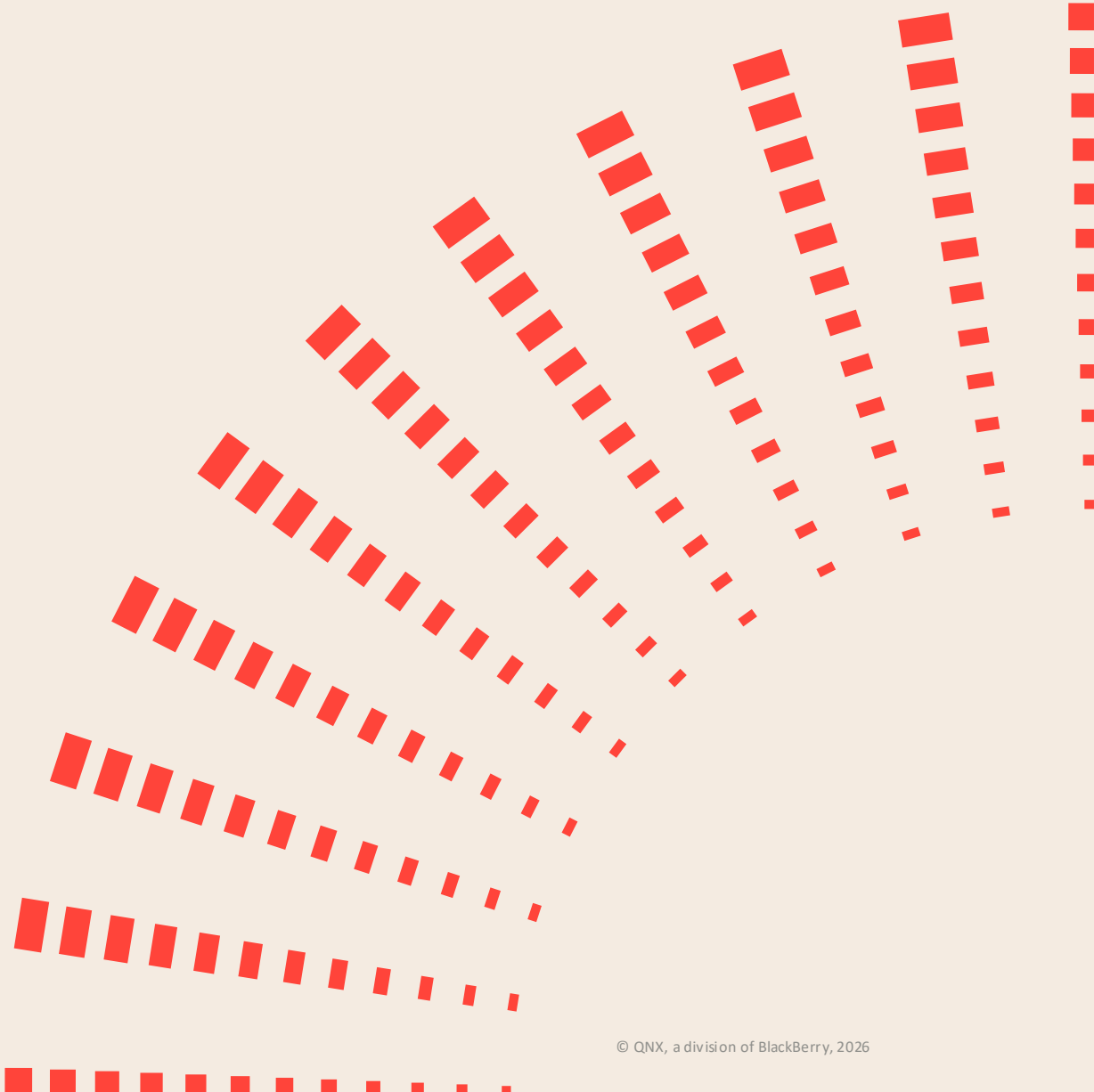
  ✂ Google “Tiger VNC”, download, install
- Distribute Pis
- Power them up, try to connect with VNC (<your-hostname.local>:5900 for VNC)
- Clone the repositories and setup camera
- Try the sample to get the camera working with a Hand Landmarking Model

Next up: Pi set up...

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QNX Project on QDD

Basma Ben Taleb, Product Management Student



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AI Hand Landmark Demo - Step-by-Step

Default username and password is qnxuser/qnxuser.

The tools `sudo` and `su` are available (root/root).

Software repository: https://github.com/qnx/QNX-AI-workshop

1. Connect to your Pi over VNC (VNC must be used to see the application)
   - Click the icon for the Desktop
   - In the Applications menu, open Terminal
2. Clone the repository we need:

   $ git clone https://github.com/qnx/QNX-AI-workshop
3. Follow the Hand Landmarking README.md to connect your camera:

   $ cd QNX-AI-workshop && nano README.md *(Ctrl+X to exit)* or read it online: https://github.com/qnx/QNX-AI-workshop/blob/main/README.md
4. Install all dependencies and ML model

   $ cd QNX-AI-workshop && sudo ./configure.sh
5. Build the demo

   $ ./build.sh
6. Try the code!

   $ ./start.sh --ros-args -p camera_unit:=5

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How to Connect to Your Raspberry Pi

Connect to the network on your laptop: use SSID "QNXLaptops", password "everywhere".

Connect to your Pi & try some commands:

- `$ ping <your-hostname>.local`
- `$ ssh qnxuser@<your-hostname>.local`

  OR

  `$ ssh qnxuser@10.20.30.40`
- **VNC Client:** `<your-hostname>.local:5900`

Default username and password is qnxuser/qnxuser.

The tools `sudo` and `su` are available (root/root).

Common QNX Commands

`# pidin` - List processes

`# slog2info` - Output logs (must be root!)

`# slay <process>`

Send a signal to p - (By default, SIGTERM)

`# use <app>` - Get usage info on app

`# top` - Get system utilization

`# hogs` - Print the system utilization



Using Windows? To SSH use the "Terminal" app and:

```
> ssh -m hmac-sha2-256 qnxuser@qnxpi00.local
```

For file transfers:

```
> scp -o MACs=hmac-sha2-256 <file> qnxuser@<your-ip>:~
```

The software repository: https://github.com/qnx/QNX-AI-workshop

FOLLOW THE README!

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Hand Landmarking AI Demo

Stretch goal

- Modify the code in `handtrack_gui.hpp` to choose a display name and color for the hand tracking. Make sure to rebuild the application and restart it to see the changes.
- Change the confidence values to see their effect on detection; see the README for all options.
- Implement basic gesture recognition from the landmark positions (thumbs up, thumbs down, open hand, closed hand).

Possible new features

- Update the interface to allow dynamically changing the machine learning configuration parameters, such as confidence values, while running with GTK4 widgets.
- Add pose landmarking with MediaPipe pose_landmarker
- Add ML gesture recognition as a next step on top of the existing models.
- Convert this project to use MediaPipe directly with our port https://github.com/qnx-ports/mediapipe/.
- Update the demo so two players can compete at Pong, using hand position as the paddles.

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Important Links

Download QNX Everywhere: https://qnx.com/getqnx

QNX for VS Code: https://www.qnx.com/developers/docs/8.0/com.qnx.doc.qnxtoolkit.user_guide/topic/quickstart_guide.html

Using QNX on a Mac: https://devblog.qnx.com/how-to-use-macos-to-develop-for-qnx-8-0/

QNX Open-source Ports: https://oss.qnx.com/

QNX GitLab: https://gitlab.com/qnx/

QNX GitHub: https://github.com/qnx/

QNX Free Online Training: https://learning.qnx.com/

QNX Discord Server: https://discord.gg/tQVpvCzD

QNX Subreddit: https://www.reddit.com/r/QNX/

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Thank you